API Specification — DSS Prioritas Perbaikan PJU & Mitigasi Kriminalitas Dini Hari Jatinangor

Versi API: 1.0.0 | **OpenAPI:** 3.1.0 | **Base URL Lokal:** `http://localhost:8000` | **Swagger UI:** `http://localhost:8000/docs`

REST API Decision Support System (DSS) Prioritas Perbaikan PJU & Mitigasi Kriminalitas Dini Hari di Kecamatan Jatinangor (Metode AHP + TOPSIS).

Autentikasi & Header Wajib (Frontend)

1. Login via `POST /api/v1/auth/login` untuk mendapatkan `access_token`.

2. Sertakan header pada seluruh endpoint terproteksi:

```
Authorization: Bearer <access_token>
```

CORS Diizinkan: `http://localhost:3000`, `http://127.0.0.1:3000`

Akun Default Seeder & Role RBAC

Username	Password	Role
`admin`	`admin123`	`ADMIN`
`operator`	`operator123`	`OPERATOR`
`penilai`	`penilai123`	`PENILAI`
`pju_lead`	`pju123`	`PENANGGUNG_JAWAB_PJU`

Aturan Bisnis Penting untuk Integrasi Frontend

- **Koridor Valid:** `"Ciseke"`, `"Caringin"`, `"Gang H Nena"`, `"Hegarmanah"`, `"Sukawening"`.
- **Geometri GeoJSON:** Mendukung `LineString` (minimal 2 titik `[lon, lat]`) dan `Polygon` (ring tertutup minimal 4 titik).
- **Batas Kedaluwarsa Inspeksi:** `90` hari (`INSPECTION_EXPIRY_DAYS`). Data melewati 90 hari ditandai `WARNING` saat CSV preview & `is_eligible: false` saat TOPSIS.
- **Konsistensi AHP:** Bobot hanya dapat diaktifkan jika `cr <= 0.10` (`is_consistent: true`).
- **Override Daftar Kerja:** Jika `set(final_selected_segment_ids) != set(system_proposed_segment_ids)`, field `override_reason` wajib minimal 10 karakter (jika tidak, API mengembalikan `HTTP 422`).
- **Kategori Peta Prioritas:** `"Tinggi"` (`>= 0.66`), `"Sedang"` (`0.33 - 0.659`), `"Rendah"` (`< 0.33`), `"Tidak Layak"` (`is_eligible=false`).

Daftar Ringkas Seluruh Endpoint (25 Endpoints)

Modul	Method	Endpoint Path	Role Diizinkan	Ringkasan
Authentication	POST	`/api/v1/auth/login`	Public (Tanpa Token)	Login
Authentication	GET	`/api/v1/auth/me`	Semua Role Login	Get Me
Users	GET	`/api/v1/users`	ADMIN	List Users
Users	POST	`/api/v1/users`	ADMIN	Create User
Users	PATCH	`/api/v1/users/{user_id}`	ADMIN	Update User
Alley Segments	GET	`/api/v1/segments`	Semua Role Login	List Segments
Alley Segments	POST	`/api/v1/segments`	OPERATOR, ADMIN	Create Segment
Alley Segments	GET	`/api/v1/segments/{segment_id}`	Semua Role Login	Get Segment Detail
Alley Segments	PUT	`/api/v1/segments/{segment_id}`	OPERATOR, ADMIN	Update Segment

Modul	Method	Endpoint Path	Role Diizinkan	Ringkasan
Observations & CSV Import	POST	`/api/v1/observations/batches/{batch_id}/confirm`	OPERATOR, ADMIN	Confirm Csv Batch
Observations & CSV Import	POST	`/api/v1/observations/form`	OPERATOR, ADMIN	Submit Observation Form
Observations & CSV Import	POST	`/api/v1/observations/upload-csv`	OPERATOR, ADMIN	Upload Csv Preview
AHP Weighting	POST	`/api/v1/ahp/calculate-preview`	PENILAI, ADMIN	Preview Ahp
AHP Weighting	GET	`/api/v1/ahp/versions`	Semua Role Login	List Ahp Versions
AHP Weighting	POST	`/api/v1/ahp/versions`	PENILAI, ADMIN	Create Ahp Version
AHP Weighting	PATCH	`/api/v1/ahp/versions/{version_id}/activate`	PENILAI, ADMIN	Activate Ahp Version
TOPSIS Ranking	POST	`/api/v1/topsis/cycles`	PENANGGUNG_JAWAB_PJU, ADMIN	Run Topsis Cycle
TOPSIS Ranking	GET	`/api/v1/topsis/cycles/{cycle_id}`	Semua Role Login	Get Topsis Cycle Detail
Priority Map	GET	`/api/v1/map/priority-layer`	Semua Role Login	Get Priority Map Layer
Priority Map	GET	`/api/v1/map/segments/{segment_id}/detail`	Semua Role Login	Get Map Segment Detail
Worklists & Override Audit	POST	`/api/v1/worklists/cycles/{cycle_id}/finalize`	PENANGGUNG_JAWAB_PJU, ADMIN	Finalize Worklist
Worklists & Override Audit	POST	`/api/v1/worklists/cycles/{cycle_id}/propose`	PENANGGUNG_JAWAB_PJU, ADMIN	Propose Worklist
History & Export	GET	`/api/v1/history/cycles`	Semua Role Login	List Cycle History
History & Export	GET	`/api/v1/history/cycles/{cycle_id}/export-csv`	Semua Role Login	Export Cycle Worklist Csv
Health	GET	`/health`	Public (Tanpa Token)	Health Check

Detail Spesifikasi Endpoint (Per Modul)

Authentication

POST `/api/v1/auth/login` Public Role: Public (Tanpa Token)

Login

Authenticate user by username and password and issue JWT access token.

REQUEST BODY (`APPLICATION/JSON`)

Field	Tipe	Status	Keterangan / Validasi
`username`	`string`	**Wajib**	minLen=1, maxLen=50
`password`	`string`	**Wajib**	minLen=1

```
{
  "username": "pju_lead",
  "password": "pju123"
}
```

RESPONSES

HTTP 200 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`access_token`	`string`	**Wajib**	
`token_type`	`string`	Opsional	default=bearer
`user`	`UserResponse`	**Wajib**	

▼ Contoh Response JSON

```
{
  "access_token": "string",
  "token_type": "bearer",
  "user": {
    "id": 1,
    "username": "string",
    "email": "user@jatinangor-dss.id",
    "full_name": "string",
    "role": "ADMIN",
    "is_active": true,
    "created_at": "2026-11-13T18:30:00Z"
  }
}
```

HTTP 422 — Validation Error

GET `/api/v1/auth/me` Bearer JWT Required Role: Semua Role Login

Get Me

Return profile of currently authenticated user.

RESPONSES

HTTP 200 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`username`	`string`	**Wajib**	

Field	Tipe	Status	Keterangan / Validasi
`email`	`string (email)`	**Wajib**	
`full_name`	`string`	**Wajib**	
`role`	`UserRole`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`created_at`	`string (date-time)`	**Wajib**	

▼ Contoh Response JSON

```
{
  "id": 1,
  "username": "string",
  "email": "user@jatinangor-dss.id",
  "full_name": "string",
  "role": "ADMIN",
  "is_active": true,
  "created_at": "2026-11-13T18:30:00Z"
}
```

Users

GET `/api/v1/users` Bearer JWT Required Role: ADMIN

List Users

Return all user accounts (ADMIN only).

RESPONSES

HTTP 200 — Successful Response

Response berupa array of objects dengan field per item:

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`username`	`string`	**Wajib**	
`email`	`string (email)`	**Wajib**	
`full_name`	`string`	**Wajib**	
`role`	`UserRole`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`created_at`	`string (date-time)`	**Wajib**	

▼ Contoh Response JSON

```
[
  {
    "id": 1,
    "username": "string",
    "email": "user@jatinangor-dss.id",
    "full_name": "string",
    "role": "ADMIN",
    "is_active": true,
    "created_at": "2026-11-13T18:30:00Z"
  }
]
```

POST `/api/v1/users` Bearer JWT Required Role: ADMIN

Create User

Create a new user account with assigned role (ADMIN only).

REQUEST BODY (`APPLICATION/JSON`)

Field	Tipe	Status	Keterangan / Validasi
`username`	`string`	**Wajib**	minLen=3, maxLen=50
`email`	`string (email)`	**Wajib**	
`full_name`	`string`	**Wajib**	minLen=2, maxLen=150
`password`	`string`	**Wajib**	minLen=6, maxLen=128
`role`	`UserRole`	**Wajib**	
`is_active`	`boolean`	Opsional	default=True

```
{
  "username": "operator2",
  "email": "operator2@jatinangor-dss.id",
  "full_name": "Operator Lapangan 2",
  "password": "password123",
  "role": "OPERATOR",
  "is_active": true
}
```

RESPONSES

HTTP 201 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`username`	`string`	**Wajib**	
`email`	`string (email)`	**Wajib**	
`full_name`	`string`	**Wajib**	
`role`	`UserRole`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`created_at`	`string (date-time)`	**Wajib**	

▼ Contoh Response JSON

```
{
  "id": 1,
  "username": "string",
  "email": "user@jatinangor-dss.id",
  "full_name": "string",
  "role": "ADMIN",
  "is_active": true,
  "created_at": "2026-11-13T18:30:00Z"
}
```

HTTP 422 — Validation Error

PATCH `/api/v1/users/{user_id}` Bearer JWT Required Role: ADMIN

Update User

Update user role, full name, or active status (ADMIN only).

PATH / QUERY PARAMETERS

Parameter	Lokasi	Tipe	Status	Deskripsi
`user_id`	path	`integer`	Wajib	

REQUEST BODY (`APPLICATION/JSON`)

Field	Tipe	Status	Keterangan / Validasi
`full_name`	`string | null`	Opsional	
`role`	`UserRole | null`	Opsional	
`is_active`	`boolean | null`	Opsional	

```
{
  "full_name": "Operator Lapangan Baru",
  "role": "OPERATOR",
  "is_active": true
}
```

RESPONSES

HTTP 200 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`username`	`string`	**Wajib**	
`email`	`string (email)`	**Wajib**	
`full_name`	`string`	**Wajib**	
`role`	`UserRole`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`created_at`	`string (date-time)`	**Wajib**	

▼ Contoh Response JSON

```
{
  "id": 1,
  "username": "string",
  "email": "user@jatinangor-dss.id",
  "full_name": "string",
  "role": "ADMIN",
  "is_active": true,
  "created_at": "2026-11-13T18:30:00Z"
}
```

HTTP 422 — Validation Error

Alley Segments

GET `/api/v1/segments` Bearer JWT Required Role: Semua Role Login

List Segments

List alley segments with optional corridor, status, and keyword filters.

PATH / QUERY PARAMETERS

Parameter	Lokasi	Tipe	Status	Deskripsi
`corridor`	query	`CorridorEnum | null`	Opsional	
`is_active`	query	`boolean | null`	Opsional	

Parameter	Lokasi	Tipe	Status	Deskripsi
`q`	query	`string | null`	Opsional	Kata kunci kode atau nama gang

RESPONSES

HTTP 200 — Successful Response

Response berupa array of objects dengan field per item:

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`segment_code`	`string`	**Wajib**	
`name`	`string`	**Wajib**	
`corridor`	`CorridorEnum`	**Wajib**	
`geojson_geometry`	`object`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`updated_at`	`string (date-time)`	**Wajib**	

▼ Contoh Response JSON

```
[
  {
    "id": 1,
    "segment_code": "string",
    "name": "string",
    "corridor": "Ciseke",
    "geojson_geometry": {
      "key": "value"
    },
    "is_active": true,
    "updated_at": "2026-11-13T18:30:00Z"
  }
]
```

HTTP 422 — Validation Error

POST `/api/v1/segments` Bearer JWT Required Role: OPERATOR, ADMIN

Create Segment

Create a new alley segment with validated GeoJSON geometry.

REQUEST BODY (`APPLICATION/JSON`)

Field	Tipe	Status	Keterangan / Validasi
`segment_code`	`string`	**Wajib**	minLen=3, maxLen=50
`name`	`string`	**Wajib**	minLen=3, maxLen=200
`corridor`	`CorridorEnum`	**Wajib**	
`geojson_geometry`	`GeoJSONGeometry`	**Wajib**	
`is_active`	`boolean`	Opsional	default=True

```
{
  "segment_code": "GG-CSK-004",
  "name": "Gang Ciseke Masjid Baru",
  "corridor": "Ciseke",
  "geojson_geometry": {
    "type": "LineString",
    "coordinates": [
      [
        107.7695,
        -6.9331
      ],
      [
        107.7706,
        -6.9339
      ]
    ]
  },
  "is_active": true
}
```


RESPONSES

HTTP 201 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`segment_code`	`string`	**Wajib**	
`name`	`string`	**Wajib**	
`corridor`	`CorridorEnum`	**Wajib**	
`geojson_geometry`	`object`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`updated_at`	`string (date-time)`	**Wajib**	

▼ Contoh Response JSON


```
{
  "id": 1,
  "segment_code": "string",
  "name": "string",
  "corridor": "Ciseke",
  "geojson_geometry": {
    "key": "value"
  },
  "is_active": true,
  "updated_at": "2026-11-13T18:30:00Z"
}
```


HTTP 422 — Validation Error

GET `/api/v1/segments/{segment_id}` Bearer JWT Required Role: Semua Role Login

Get Segment Detail

Return detail of a single alley segment along with its latest observation.

PATH / QUERY PARAMETERS

Parameter	Lokasi	Tipe	Status	Deskripsi
`segment_id`	path	`integer`	Wajib	

RESPONSES

HTTP 200 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`segment_code`	`string`	**Wajib**	
`name`	`string`	**Wajib**	
`corridor`	`CorridorEnum`	**Wajib**	
`geojson_geometry`	`object`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`updated_at`	`string (date-time)`	**Wajib**	
`observation`	`ObservationBriefResponse | null`	Opsional	

▼ Contoh Response JSON


```
{
  "id": 1,
  "segment_code": "string",
  "name": "string",
  "corridor": "Ciseke",
  "geojson_geometry": {
    "key": "value"
  },
  "is_active": true,
  "updated_at": "2026-11-13T18:30:00Z",
  "observation": {
    "id": 1,
    "segment_id": 1,
    "total_lights": 1,
    "broken_lights": 1,
    "dawn_crime_count": 1,
    "boarding_house_count": 1,
    "inspection_date": "2026-11-13",
    "data_source": "MANUAL_FORM",
    "updated_by": 1,
    "updated_at": "2026-11-13T18:30:00Z"
  }
}
```


HTTP 422 — Validation Error

PUT `/api/v1/segments/{segment_id}` Bearer JWT Required Role: OPERATOR, ADMIN

Update Segment

Update alley segment metadata or GeoJSON geometry.

PATH / QUERY PARAMETERS

Parameter	Lokasi	Tipe	Status	Deskripsi
`segment_id`	path	`integer`	Wajib	

REQUEST BODY (`APPLICATION/JSON`)

Field	Tipe	Status	Keterangan / Validasi
`name`	`string | null`	Opsional	
`corridor`	`CorridorEnum | null`	Opsional	
`geojson_geometry`	`GeoJSONGeometry | null`	Opsional	
`is_active`	`boolean | null`	Opsional	

```
{
  "name": "Gang Ciseke Lama Utara (Revisi)",
  "is_active": true
}
```

RESPONSES

HTTP 200 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`segment_code`	`string`	**Wajib**	
`name`	`string`	**Wajib**	
`corridor`	`CorridorEnum`	**Wajib**	
`geojson_geometry`	`object`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`updated_at`	`string (date-time)`	**Wajib**	

▼ Contoh Response JSON

```
{
  "id": 1,
  "segment_code": "string",
  "name": "string",
  "corridor": "Ciseke",
  "geojson_geometry": {
    "key": "value"
  },
  "is_active": true,
  "updated_at": "2026-11-13T18:30:00Z"
}
```

HTTP 422 — Validation Error

Observations & CSV Import

POST `/api/v1/observations/batches/{batch_id}/confirm` Bearer JWT Required Role: OPERATOR, ADMIN

Confirm Csv Batch

Confirm and commit valid/corrected rows from staging into segment_observations.

PATH / QUERY PARAMETERS

Parameter	Lokasi	Tipe	Status	Deskripsi
`batch_id`	path	`string`	Wajib	

REQUEST BODY (`APPLICATION/JSON`)

Field	Tipe	Status	Keterangan / Validasi
`include_warnings`	`boolean`	Opsional	default=True
`corrections`	`array[CSVRowCorrection]`	Opsional	

```
{
  "include_warnings": true,
  "corrections": []
}
```

RESPONSES

HTTP 200 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`batch_id`	`string`	**Wajib**	
`status`	`BatchStatus`	**Wajib**	
`committed_count`	`integer`	**Wajib**	
`skipped_error_count`	`integer`	**Wajib**	
`committed_segment_codes`	`array[string]`	**Wajib**	

▼ Contoh Response JSON

```
{
  "batch_id": "string",
  "status": "PENDING_CONFIRMATION",
  "committed_count": 1,
  "skipped_error_count": 1,
  "committed_segment_codes": [
    "string"
  ]
}
```

HTTP 422 — Validation Error

POST `/api/v1/observations/form` Bearer JWT Required Role: OPERATOR, ADMIN

Submit Observation Form

Create or update a segment observation via manual form with validation checks.

REQUEST BODY (`APPLICATION/JSON`)

Field	Tipe	Status	Keterangan / Validasi
`segment_id`	`integer | null`	Opsional	
`segment_code`	`string | null`	Opsional	
`total_lights`	`integer`	**Wajib**	min=0.0
`broken_lights`	`integer`	**Wajib**	min=0.0
`dawn_crime_count`	`integer | null`	Opsional	
`boarding_house_count`	`integer`	**Wajib**	min=0.0
`inspection_date`	`string (date)`	**Wajib**	

```
{
  "segment_code": "GG-CSK-001",
  "total_lights": 12,
  "broken_lights": 7,
  "dawn_crime_count": 5,
  "boarding_house_count": 18,
  "inspection_date": "2026-11-10"
}
```

RESPONSES

HTTP 200 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`observation`	`ObservationBriefResponse`	**Wajib**	
`warnings`	`array[string]`	Opsional	

▼ Contoh Response JSON

```
{
  "observation": {
    "id": 1,
    "segment_id": 1,
    "total_lights": 1,
    "broken_lights": 1,
    "dawn_crime_count": 1,
    "boarding_house_count": 1,
    "inspection_date": "2026-11-13",
    "data_source": "MANUAL_FORM",
    "updated_by": 1,
    "updated_at": "2026-11-13T18:30:00Z"
  },
  "warnings": [
    "string"
  ]
}
```


HTTP 422 — Validation Error

POST `/api/v1/observations/upload-csv` Bearer JWT Required Role: OPERATOR, ADMIN

Upload Csv Preview

Upload a CSV file, run pre-commit validations, and store a staging batch.

REQUEST BODY (`MULTIPART/FORM-DATA`)

Field	Tipe	Status	Keterangan / Validasi
`file`	`string`	**Wajib**	

Gunakan `FormData` dengan field `file` berisi file `.csv` berkolom:
`segment_code,total_lights,broken_lights,dawn_crime_count,boarding_house_count,inspection_date`.

RESPONSES

HTTP 201 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`batch_id`	`string`	**Wajib**	
`filename`	`string`	**Wajib**	
`status`	`BatchStatus`	**Wajib**	
`total_rows`	`integer`	**Wajib**	
`valid_rows`	`integer`	**Wajib**	
`warning_rows`	`integer`	**Wajib**	
`error_rows`	`integer`	**Wajib**	
`rows`	`array[CSVValidatedRow]`	**Wajib**	
`created_at`	`string (date-time)`	**Wajib**	

▼ Contoh Response JSON

```
{
  "batch_id": "string",
  "filename": "string",
  "status": "PENDING_CONFIRMATION",
  "total_rows": 1,
  "valid_rows": 1,
  "warning_rows": 1,
  "error_rows": 1,
  "rows": [
    {
      "row_number": 1,
      "segment_code": "string",
      "total_lights": 1,
      "broken_lights": 1,
      "dawn_crime_count": 1,
      "boarding_house_count": 1,
      "inspection_date": "string",
      "status": "VALID",
      "issues": [
        {
          "code": "string",
          "severity": "WARNING",
          "message": "string"
        }
      ],
      "messages": [
        "string"
      ]
    }
  ],
  "created_at": "2026-11-13T18:30:00Z"
}
```


HTTP 422 — Validation Error

AHP Weighting

POST `/api/v1/ahp/calculate-preview` Bearer JWT Required Role: PENILAI, ADMIN

Preview Ahp

Calculate AHP priority weights and CR without persisting to database.

REQUEST BODY (`APPLICATION/JSON`)

Field	Tipe	Status	Keterangan / Validasi
`pairwise_matrix`	`array[array[number]]`	**Wajib**	Matriks perbandingan berpasangan 3x3 untuk [C1: Kerusakan Lampu, C2: Curanmor Dini Hari, C3: Kepadatan Indekos].

```
{
  "pairwise_matrix": [
    [
      1.0,
      0.5,
      2.0
    ],
    [
      2.0,
      1.0,
      3.0
    ],
    [
      0.5,
      0.333333,
      1.0
    ]
  ]
}
```


RESPONSES

HTTP 200 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`pairwise_matrix`	`array[array[number]]`	**Wajib**	
`w_light_damage`	`number`	**Wajib**	
`w_dawn_crime`	`number`	**Wajib**	
`w_boarding_density`	`number`	**Wajib**	
`lambda_max`	`number`	**Wajib**	
`ci`	`number`	**Wajib**	
`ri`	`number`	**Wajib**	
`cr`	`number`	**Wajib**	
`is_consistent`	`boolean`	**Wajib**	

▼ Contoh Response JSON


```
{
  "pairwise_matrix": [
    [
      0.5
    ]
  ],
  "w_light_damage": 0.5,
  "w_dawn_crime": 0.5,
  "w_boarding_density": 0.5,
  "lambda_max": 0.5,
  "ci": 0.5,
  "ri": 0.5,
  "cr": 0.5,
  "is_consistent": true
}
```


HTTP 422 — Validation Error

GET `/api/v1/ahp/versions` Bearer JWT Required Role: Semua Role Login

List Ahp Versions

List all saved AHP weight versions ordered from newest to oldest.

RESPONSES

HTTP 200 — Successful Response

Response berupa array of objects dengan field per item:

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`version_name`	`string`	**Wajib**	
`created_by`	`integer`	**Wajib**	
`pairwise_matrix`	`array[array[number]]`	**Wajib**	
`w_light_damage`	`number`	**Wajib**	
`w_dawn_crime`	`number`	**Wajib**	
`w_boarding_density`	`number`	**Wajib**	
`lambda_max`	`number`	**Wajib**	
`ci`	`number`	**Wajib**	
`ri`	`number`	**Wajib**	
`cr`	`number`	**Wajib**	
`is_consistent`	`boolean`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`created_at`	`string (date-time)`	**Wajib**	

▼ Contoh Response JSON


```
[
  {
    "id": 1,
    "version_name": "string",
    "created_by": 1,
    "pairwise_matrix": [
      [
        0.5
      ]
    ],
    "w_light_damage": 0.5,
    "w_dawn_crime": 0.5,
    "w_boarding_density": 0.5,
    "lambda_max": 0.5,
    "ci": 0.5,
    "ri": 0.5,
    "cr": 0.5,
    "is_consistent": true,
    "is_active": true,
    "created_at": "2026-11-13T18:30:00Z"
  }
]
```


POST `/api/v1/ahp/versions` Bearer JWT Required Role: PENILAI, ADMIN

Create Ahp Version

Calculate and store an AHP weight version; activate only if CR <= 0.10.

REQUEST BODY (`APPLICATION/JSON`)

Field	Tipe	Status	Keterangan / Validasi
`pairwise_matrix`	`array[array[number]]`	**Wajib**	Matriks perbandingan berpasangan 3x3 untuk [C1: Kerusakan Lampu, C2: Curanmor Dini Hari, C3: Kepadatan Indekos].

Field	Tipe	Status	Keterangan / Validasi
`version_name`	`string`	**Wajib**	minLen=2, maxLen=150
`activate_immediately`	`boolean`	Opsional	default=False


```
{
  "version_name": "Bobot Pakar Q4 2026",
  "pairwise_matrix": [
    [
      1.0,
      0.5,
      2.0
    ],
    [
      2.0,
      1.0,
      3.0
    ],
    [
      0.5,
      0.333333,
      1.0
    ]
  ],
  "activate_immediately": true
}
```


RESPONSES

HTTP 201 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`version_name`	`string`	**Wajib**	
`created_by`	`integer`	**Wajib**	
`pairwise_matrix`	`array[array[number]]`	**Wajib**	
`w_light_damage`	`number`	**Wajib**	
`w_dawn_crime`	`number`	**Wajib**	
`w_boarding_density`	`number`	**Wajib**	
`lambda_max`	`number`	**Wajib**	
`ci`	`number`	**Wajib**	
`ri`	`number`	**Wajib**	
`cr`	`number`	**Wajib**	
`is_consistent`	`boolean`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`created_at`	`string (date-time)`	**Wajib**	

▼ Contoh Response JSON

```
{
  "id": 1,
  "version_name": "string",
  "created_by": 1,
  "pairwise_matrix": [
    [
      0.5
    ]
  ],
  "w_light_damage": 0.5,
  "w_dawn_crime": 0.5,
  "w_boarding_density": 0.5,
  "lambda_max": 0.5,
  "ci": 0.5,
  "ri": 0.5,
  "cr": 0.5,
  "is_consistent": true,
  "is_active": true,
  "created_at": "2026-11-13T18:30:00Z"
}
```


HTTP 422 — Validation Error

PATCH `/api/v1/ahp/versions/{version_id}/activate` Bearer JWT Required Role: PENILAI, ADMIN

Activate Ahp Version

Activate an existing consistent AHP weight version (rejects CR > 0.10 with 400).

PATH / QUERY PARAMETERS

Parameter	Lokasi	Tipe	Status	Deskripsi
`version_id`	path	`integer`	Wajib	

RESPONSES

HTTP 200 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`version_name`	`string`	**Wajib**	
`created_by`	`integer`	**Wajib**	
`pairwise_matrix`	`array[array[number]]`	**Wajib**	
`w_light_damage`	`number`	**Wajib**	
`w_dawn_crime`	`number`	**Wajib**	
`w_boarding_density`	`number`	**Wajib**	
`lambda_max`	`number`	**Wajib**	
`ci`	`number`	**Wajib**	
`ri`	`number`	**Wajib**	
`cr`	`number`	**Wajib**	
`is_consistent`	`boolean`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`created_at`	`string (date-time)`	**Wajib**	

▼ Contoh Response JSON

```
{
  "id": 1,
  "version_name": "string",
  "created_by": 1,
  "pairwise_matrix": [
    [
      0.5
    ]
  ],
  "w_light_damage": 0.5,
  "w_dawn_crime": 0.5,
  "w_boarding_density": 0.5,
  "lambda_max": 0.5,
  "ci": 0.5,
  "ri": 0.5,
  "cr": 0.5,
  "is_consistent": true,
  "is_active": true,
  "created_at": "2026-11-13T18:30:00Z"
}
```


HTTP 422 — Validation Error

TOPSIS Ranking

POST `/api/v1/topsis/cycles` Bearer JWT Required Role: PENANGGUNG_JAWAB_PJU, ADMIN

Run Topsis Cycle

Execute a new TOPSIS ranking cycle using the active consistent AHP weight version.

REQUEST BODY (APPLICATION/JSON)

Field	Tipe	Status	Keterangan / Validasi
`cycle_name`	`string`	**Wajib**	minLen=2, maxLen=200
`criteria_directions`	`array[any]`	Opsional	default=['max', 'max', 'max']


```
{
  "cycle_name": "Siklus Evaluasi November 2026",
  "criteria_directions": [
    "max",
    "max",
    "max"
  ]
}
```


RESPONSES

HTTP 201 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`cycle_id`	`integer`	**Wajib**	
`cycle_name`	`string`	**Wajib**	
`executed_by`	`integer`	**Wajib**	
`ahp_version_id`	`integer`	**Wajib**	
`weights_snapshot`	`object`	**Wajib**	
`status`	`CycleStatus`	**Wajib**	
`created_at`	`string (date-time)`	**Wajib**	
`eligible_count`	`integer`	**Wajib**	

Field	Tipe	Status	Keterangan / Validasi
`ineligible_count`	`integer`	**Wajib**	
`eligible_results`	`array[CycleRankingItemResponse]`	**Wajib**	
`ineligible_results`	`array[CycleRankingItemResponse]`	**Wajib**	

▼ Contoh Response JSON

```
{
  "cycle_id": 1,
  "cycle_name": "string",
  "executed_by": 1,
  "ahp_version_id": 1,
  "weights_snapshot": {
    "key": "value"
  },
  "status": "RANKED",
  "created_at": "2026-11-13T18:30:00Z",
  "eligible_count": 1,
  "ineligible_count": 1,
  "eligible_results": [
    {
      "id": 1,
      "segment_id": 1,
      "segment_code": "string",
      "segment_name": "string",
      "corridor": "Ciseke",
      "is_eligible": true,
      "ineligible_reason": "string",
      "snapshot_data": {
        "key": "value"
      },
      "norm_c1": 0.5,
      "norm_c2": 0.5,
      "norm_c3": 0.5,
      "weighted_c1": 0.5,
      "weighted_c2": 0.5,
      "weighted_c3": 0.5,
      "d_plus": 0.5,
      "d_minus": 0.5,
      "preference_score": 0.5,
      "rank": 1,
      "priority_category": "string",
      "is_tie_broken": true
    }
  ],
  "ineligible_results": [
    {
      "id": 1,
      "segment_id": 1,
      "segment_code": "string",
      "segment_name": "string",
      "corridor": "Ciseke",
      "is_eligible": true,
      "ineligible_reason": "string",
      "snapshot_data": {
        "key": "value"
      },
      "norm_c1": 0.5,
      "norm_c2": 0.5,
      "norm_c3": 0.5,
      "weighted_c1": 0.5,
      "weighted_c2": 0.5,
      "weighted_c3": 0.5,
      "d_plus": 0.5,
      "d_minus": 0.5,
      "preference_score": 0.5,
      "rank": 1,
      "priority_category": "string",
      "is_tie_broken": true
    }
  ]
}
```


HTTP 422 — Validation Error

GET `/api/v1/topsis/cycles/{cycle_id}` Bearer JWT Required Role: Semua Role Login

Get Topsis Cycle Detail

Retrieve ranked eligible segments and ineligible segments for a cycle.

PATH / QUERY PARAMETERS

Parameter	Lokasi	Tipe	Status	Deskripsi
`cycle_id`	path	`integer`	Wajib	

RESPONSES

HTTP 200 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`cycle_id`	`integer`	**Wajib**	
`cycle_name`	`string`	**Wajib**	
`executed_by`	`integer`	**Wajib**	
`ahp_version_id`	`integer`	**Wajib**	
`weights_snapshot`	`object`	**Wajib**	
`status`	`CycleStatus`	**Wajib**	
`created_at`	`string (date-time)`	**Wajib**	
`eligible_count`	`integer`	**Wajib**	
`ineligible_count`	`integer`	**Wajib**	
`eligible_results`	`array[CycleRankingItemResponse]`	**Wajib**	
`ineligible_results`	`array[CycleRankingItemResponse]`	**Wajib**	

▼ Contoh Response JSON

```
{
  "cycle_id": 1,
  "cycle_name": "string",
  "executed_by": 1,
  "ahp_version_id": 1,
  "weights_snapshot": {
    "key": "value"
  },
  "status": "RANKED",
  "created_at": "2026-11-13T18:30:00Z",
  "eligible_count": 1,
  "ineligible_count": 1,
  "eligible_results": [
    {
      "id": 1,
      "segment_id": 1,
      "segment_code": "string",
      "segment_name": "string",
      "corridor": "Ciseke",
      "is_eligible": true,
      "ineligible_reason": "string",
      "snapshot_data": {
        "key": "value"
      },
      "norm_c1": 0.5,
      "norm_c2": 0.5,
      "norm_c3": 0.5,
      "weighted_c1": 0.5,
      "weighted_c2": 0.5,
      "weighted_c3": 0.5,
      "d_plus": 0.5,
      "d_minus": 0.5,
      "preference_score": 0.5,
      "rank": 1,
      "priority_category": "string",
      "is_tie_broken": true
    }
  ],
  "ineligible_results": [
    {
      "id": 1,
      "segment_id": 1,
      "segment_code": "string",
      "segment_name": "string",
      "corridor": "Ciseke",
      "is_eligible": true,
      "ineligible_reason": "string",
      "snapshot_data": {
        "key": "value"
      },
      "norm_c1": 0.5,
      "norm_c2": 0.5,
      "norm_c3": 0.5,
      "weighted_c1": 0.5,
      "weighted_c2": 0.5,
      "weighted_c3": 0.5,
      "d_plus": 0.5,
      "d_minus": 0.5,
      "preference_score": 0.5,
      "rank": 1,
      "priority_category": "string",
      "is_tie_broken": true
    }
  ]
}
```


HTTP 422 — Validation Error

Priority Map

GET `/api/v1/map/priority-layer` Bearer JWT Required Role: Semua Role Login

Get Priority Map Layer

Return a GeoJSON FeatureCollection of ranked alley segments for map rendering.

PATH / QUERY PARAMETERS

Parameter	Lokasi	Tipe	Status	Deskripsi
`cycle_id`	query	`integer | null`	Opsional	
`corridor`	query	`CorridorEnum | null`	Opsional	

RESPONSES

HTTP 200 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`type`	`string`	Opsional	default=FeatureCollection
`cycle_id`	`integer | null`	Opsional	
`cycle_name`	`string | null`	Opsional	
`features`	`array[GeoJSONFeature]`	**Wajib**	

▼ Contoh Response JSON

```
{
  "type": "FeatureCollection",
  "cycle_id": 1,
  "cycle_name": "string",
  "features": [
    {
      "type": "Feature",
      "geometry": {
        "key": "value"
      },
      "properties": {
        "key": "value"
      }
    }
  ]
}
```

HTTP 422 — Validation Error

GET `/api/v1/map/segments/{segment_id}/detail` Bearer JWT Required Role: Semua Role Login

Get Map Segment Detail

Return full segment attributes, TOPSIS computation breakdown, and data provenance.

PATH / QUERY PARAMETERS

Parameter	Lokasi	Tipe	Status	Deskripsi
`segment_id`	path	`integer`	Wajib	
`cycle_id`	query	`integer | null`	Opsional	

RESPONSES

HTTP 200 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`segment_id`	`integer`	**Wajib**	

Field	Tipe	Status	Keterangan / Validasi
`segment_code`	`string`	**Wajib**	
`name`	`string`	**Wajib**	
`corridor`	`CorridorEnum`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`geojson_geometry`	`object`	**Wajib**	
`cycle_id`	`integer | null`	Opsional	
`cycle_name`	`string | null`	Opsional	
`ahp_weights`	`object | null`	Opsional	
`indicators`	`object | null`	Opsional	
`normalized_values`	`object | null`	Opsional	
`weighted_values`	`object | null`	Opsional	
`d_plus`	`number | null`	Opsional	
`d_minus`	`number | null`	Opsional	
`preference_score`	`number | null`	Opsional	
`rank`	`integer | null`	Opsional	
`priority_category`	`string`	**Wajib**	
`is_eligible`	`boolean | null`	Opsional	
`ineligible_reason`	`string | null`	Opsional	
`is_tie_broken`	`boolean`	Opsional	default=False
`inspection_date`	`string (date) | string | null`	Opsional	
`data_source`	`string | null`	Opsional	

▼ Contoh Response JSON

```
{
  "segment_id": 1,
  "segment_code": "string",
  "name": "string",
  "corridor": "Ciseke",
  "is_active": true,
  "geojson_geometry": {
    "key": "value"
  },
  "cycle_id": 1,
  "cycle_name": "string",
  "ahp_weights": {
    "key": "value"
  },
  "indicators": {
    "key": "value"
  },
  "normalized_values": {
    "key": "value"
  },
  "weighted_values": {
    "key": "value"
  },
  "d_plus": 0.5,
  "d_minus": 0.5,
  "preference_score": 0.5,
  "rank": 1,
  "priority_category": "string",
  "is_eligible": true,
  "ineligible_reason": "string",
  "is_tie_broken": false,
  "inspection_date": "2026-11-13",
  "data_source": "string"
}
```


HTTP 422 — Validation Error

Worklists & Override Audit

POST `/api/v1/worklists/cycles/{cycle_id}/finalize` Bearer JWT Required

Role: PENANGGUNG_JAWAB_PJU, ADMIN

Finalize Worklist

Finalize worklist selection, enforcing mandatory override justification.

PATH / QUERY PARAMETERS

Parameter	Lokasi	Tipe	Status	Deskripsi
`cycle_id`	path	`integer`	Wajib	

REQUEST BODY (`APPLICATION/JSON`)

Field	Tipe	Status	Keterangan / Validasi
`capacity_quota`	`integer`	**Wajib**	min > 0.0
`final_selected_segment_ids`	`array[integer]`	**Wajib**	
`override_reason`	`string | null`	Opsional	

```
{
  "capacity_quota": 3,
  "final_selected_segment_ids": [
    1,
    4,
    6
  ],
  "override_reason": "Segmen 6 memiliki akses utama menuju area kos mahasiswa yang rawan."
}
```

RESPONSES

HTTP 201 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`cycle_id`	`integer`	**Wajib**	
`decided_by`	`integer`	**Wajib**	
`capacity_quota`	`integer`	**Wajib**	
`system_proposed_segment_ids`	`array[integer]`	**Wajib**	
`final_selected_segment_ids`	`array[integer]`	**Wajib**	
`is_overridden`	`boolean`	**Wajib**	
`override_reason`	`string | null`	**Wajib**	
`finalized_at`	`string (date-time)`	**Wajib**	

▼ Contoh Response JSON

```
{
  "id": 1,
  "cycle_id": 1,
  "decided_by": 1,
  "capacity_quota": 1,
  "system_proposed_segment_ids": [
    1
  ],
  "final_selected_segment_ids": [
    1
  ],
  "is_overridden": true,
  "override_reason": "string",
  "finalized_at": "2026-11-13T18:30:00Z"
}
```

HTTP 422 — Validation Error

POST `/api/v1/worklists/cycles/{cycle_id}/propose` Bearer JWT Required

Role: PENANGGUNG_JAWAB_PJU, ADMIN

Propose Worklist

Return Top-K system-proposed segments and reserve eligible candidates.

PATH / QUERY PARAMETERS

Parameter	Lokasi	Tipe	Status	Deskripsi
`cycle_id`	path	`integer`	Wajib	

REQUEST BODY (`APPLICATION/JSON`)

Field	Tipe	Status	Keterangan / Validasi
`capacity_quota`	`integer`	**Wajib**	Kapasitas jumlah gang K > 0 — min > 0.0

```
{
  "capacity_quota": 5
}
```

RESPONSES

HTTP 200 — Successful Response

Field	Tipe	Status	Keterangan / Validasi
`cycle_id`	`integer`	**Wajib**	
`cycle_name`	`string`	**Wajib**	
`capacity_quota`	`integer`	**Wajib**	
`total_eligible`	`integer`	**Wajib**	
`system_proposed`	`array[WorklistCandidateItem]`	**Wajib**	
`reserve_candidates`	`array[WorklistCandidateItem]`	**Wajib**	

▼ Contoh Response JSON

```
{
  "cycle_id": 1,
  "cycle_name": "string",
  "capacity_quota": 1,
  "total_eligible": 1,
  "system_proposed": [
    {
      "segment_id": 1,
      "segment_code": "string",
      "segment_name": "string",
      "corridor": "Ciseke",
      "rank": 1,
      "preference_score": 0.5,
      "broken_lights": 1,
      "dawn_crime_count": 1,
      "boarding_house_count": 1,
      "is_tie_broken": true
    }
  ],
  "reserve_candidates": [
    {
      "segment_id": 1,
      "segment_code": "string",
      "segment_name": "string",
      "corridor": "Ciseke",
      "rank": 1,
      "preference_score": 0.5,
      "broken_lights": 1,
      "dawn_crime_count": 1,
      "boarding_house_count": 1,
      "is_tie_broken": true
    }
  ]
}
```

HTTP 422 — Validation Error

History & Export

GET `/api/v1/history/cycles` Bearer JWT Required Role: Semua Role Login

List Cycle History

List historical TOPSIS evaluation cycles with worklist and override summary.

RESPONSES

HTTP 200 — Successful Response

Response berupa array of objects dengan field per item:

Field	Tipe	Status	Keterangan / Validasi
`cycle_id`	`integer`	**Wajib**	
`cycle_name`	`string`	**Wajib**	
`created_at`	`string (date-time)`	**Wajib**	
`executed_by`	`integer`	**Wajib**	
`executed_by_username`	`string`	**Wajib**	
`ahp_version_id`	`integer`	**Wajib**	
`ahp_version_name`	`string`	**Wajib**	
`weights_snapshot`	`object`	**Wajib**	
`status`	`CycleStatus`	**Wajib**	
`total_segments`	`integer`	**Wajib**	
`eligible_count`	`integer`	**Wajib**	
`ineligible_count`	`integer`	**Wajib**	
`is_overridden`	`boolean | null`	Opsional	
`capacity_quota`	`integer | null`	Opsional	
`finalized_at`	`string (date-time) | null`	Opsional	

▼ Contoh Response JSON


```
[
  {
    "cycle_id": 1,
    "cycle_name": "string",
    "created_at": "2026-11-13T18:30:00Z",
    "executed_by": 1,
    "executed_by_username": "string",
    "ahp_version_id": 1,
    "ahp_version_name": "string",
    "weights_snapshot": {
      "key": "value"
    },
    "status": "RANKED",
    "total_segments": 1,
    "eligible_count": 1,
    "ineligible_count": 1,
    "is_overridden": true,
    "capacity_quota": 1,
    "finalized_at": "2026-11-13T18:30:00Z"
  }
]
```

GET `/api/v1/history/cycles/{cycle_id}/export-csv` Bearer JWT Required Role: Semua Role Login

Export Cycle Worklist Csv

Export the finalized repair worklist of an evaluation cycle to CSV.

PATH / QUERY PARAMETERS

Parameter	Lokasi	Tipe	Status	Deskripsi
`cycle_id`	path	`integer`	Wajib	

RESPONSES

HTTP 200 — Successful Response

Tipe: `any`

▼ Contoh Response JSON

```
null
```

HTTP 422 — Validation Error

Health

GET `/health` Public Role: Public (Tanpa Token)

Health Check

Simple healthcheck endpoint.

RESPONSES

HTTP 200 — Successful Response

Tipe: `object`

▼ Contoh Response JSON

```
{
  "key": "value"
}
```

Lampiran: Skema Model & Enum (TypeScript / Pydantic Reference)

AHPCalculateRequest

Field	Tipe	Status	Keterangan / Validasi
`pairwise_matrix`	`array[array[number]]`	**Wajib**	Matriks perbandingan berpasangan 3x3 untuk [C1: Kerusakan Lampu, C2: Curanmor Dini Hari, C3: Kepadatan Indekos].

AHPCalculateResponse

Field	Tipe	Status	Keterangan / Validasi
`pairwise_matrix`	`array[array[number]]`	**Wajib**	
`w_light_damage`	`number`	**Wajib**	
`w_dawn_crime`	`number`	**Wajib**	
`w_boarding_density`	`number`	**Wajib**	
`lambda_max`	`number`	**Wajib**	
`ci`	`number`	**Wajib**	
`ri`	`number`	**Wajib**	
`cr`	`number`	**Wajib**	
`is_consistent`	`boolean`	**Wajib**	

AHPVersionCreateRequest

Field	Tipe	Status	Keterangan / Validasi
`pairwise_matrix`	`array[array[number]]`	**Wajib**	Matriks perbandingan berpasangan 3x3 untuk [C1: Kerusakan Lampu, C2: Curanmor Dini Hari, C3: Kepadatan Indekos].
`version_name`	`string`	**Wajib**	minLen=2, maxLen=150
`activate_immediately`	`boolean`	Opsional	default=False

AHPVersionResponse

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`version_name`	`string`	**Wajib**	
`created_by`	`integer`	**Wajib**	
`pairwise_matrix`	`array[array[number]]`	**Wajib**	
`w_light_damage`	`number`	**Wajib**	
`w_dawn_crime`	`number`	**Wajib**	
`w_boarding_density`	`number`	**Wajib**	
`lambda_max`	`number`	**Wajib**	
`ci`	`number`	**Wajib**	
`ri`	`number`	**Wajib**	
`cr`	`number`	**Wajib**	
`is_consistent`	`boolean`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`created_at`	`string (date-time)`	**Wajib**	

BatchStatus

Enum values: `PENDING_CONFIRMATION`, `CONFIRMED`, `CANCELLED`

CSVConfirmRequest

Field	Tipe	Status	Keterangan / Validasi
`include_warnings`	`boolean`	Opsional	default=True
`corrections`	`array[CSVRowCorrection]`	Opsional	

CSVConfirmResponse

Field	Tipe	Status	Keterangan / Validasi
`batch_id`	`string`	**Wajib**	
`status`	`BatchStatus`	**Wajib**	
`committed_count`	`integer`	**Wajib**	
`skipped_error_count`	`integer`	**Wajib**	
`committed_segment_codes`	`array[string]`	**Wajib**	

CSVRowCorrection

Field	Tipe	Status	Keterangan / Validasi
`row_number`	`integer`	**Wajib**	min=1.0
`segment_code`	`string`	**Wajib**	minLen=1
`total_lights`	`integer`	**Wajib**	min=0.0
`broken_lights`	`integer`	**Wajib**	min=0.0
`dawn_crime_count`	`integer | null`	Opsional	
`boarding_house_count`	`integer`	**Wajib**	min=0.0
`inspection_date`	`string (date)`	**Wajib**	

CSVUploadPreviewResponse

Field	Tipe	Status	Keterangan / Validasi
`batch_id`	`string`	**Wajib**	
`filename`	`string`	**Wajib**	
`status`	`BatchStatus`	**Wajib**	
`total_rows`	`integer`	**Wajib**	
`valid_rows`	`integer`	**Wajib**	
`warning_rows`	`integer`	**Wajib**	
`error_rows`	`integer`	**Wajib**	
`rows`	`array[CSVValidatedRow]`	**Wajib**	
`created_at`	`string (date-time)`	**Wajib**	

CSVValidatedRow

Field	Tipe	Status	Keterangan / Validasi
`row_number`	`integer`	**Wajib**	
`segment_code`	`string`	**Wajib**	
`total_lights`	`integer | null`	Opsional	
`broken_lights`	`integer | null`	Opsional	
`dawn_crime_count`	`integer | null`	Opsional	
`boarding_house_count`	`integer | null`	Opsional	
`inspection_date`	`string | null`	Opsional	
`status`	`string (enum: 'VALID', 'WARNING', 'ERROR')`	**Wajib**	
`issues`	`array[CSVValidationIssue]`	Opsional	
`messages`	`array[string]`	Opsional	

CSVValidationIssue

Field	Tipe	Status	Keterangan / Validasi
`code`	`string`	**Wajib**	
`severity`	`string (enum: 'WARNING', 'ERROR')`	**Wajib**	
`message`	`string`	**Wajib**	

CorridorEnum

Enum values: `Ciseke`, `Caringin`, `Gang H Nena`, `Hegarmanah`, `Sukawening`

CycleHistorySummaryResponse

Field	Tipe	Status	Keterangan / Validasi
`cycle_id`	`integer`	**Wajib**	
`cycle_name`	`string`	**Wajib**	
`created_at`	`string (date-time)`	**Wajib**	
`executed_by`	`integer`	**Wajib**	
`executed_by_username`	`string`	**Wajib**	
`ahp_version_id`	`integer`	**Wajib**	
`ahp_version_name`	`string`	**Wajib**	
`weights_snapshot`	`object`	**Wajib**	
`status`	`CycleStatus`	**Wajib**	
`total_segments`	`integer`	**Wajib**	
`eligible_count`	`integer`	**Wajib**	
`ineligible_count`	`integer`	**Wajib**	
`is_overridden`	`boolean | null`	Opsional	
`capacity_quota`	`integer | null`	Opsional	
`finalized_at`	`string (date-time) | null`	Opsional	

CycleRankingItemResponse

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`segment_id`	`integer`	**Wajib**	
`segment_code`	`string`	**Wajib**	
`segment_name`	`string`	**Wajib**	
`corridor`	`CorridorEnum`	**Wajib**	
`is_eligible`	`boolean`	**Wajib**	
`ineligible_reason`	`string | null`	**Wajib**	
`snapshot_data`	`object`	**Wajib**	
`norm_c1`	`number | null`	**Wajib**	
`norm_c2`	`number | null`	**Wajib**	
`norm_c3`	`number | null`	**Wajib**	
`weighted_c1`	`number | null`	**Wajib**	
`weighted_c2`	`number | null`	**Wajib**	
`weighted_c3`	`number | null`	**Wajib**	
`d_plus`	`number | null`	**Wajib**	
`d_minus`	`number | null`	**Wajib**	
`preference_score`	`number | null`	**Wajib**	
`rank`	`integer | null`	**Wajib**	
`priority_category`	`string`	**Wajib**	
`is_tie_broken`	`boolean`	**Wajib**	

CycleStatus

Enum values: `RANKED`, `WORKLIST_FINALIZED`

DataSourceEnum

Enum values: `MANUAL_FORM`, `CSV_IMPORT`

GeoJSONFeature

Field	Tipe	Status	Keterangan / Validasi
`type`	`string`	Opsional	default=Feature
`geometry`	`object`	**Wajib**	
`properties`	`object`	**Wajib**	

GeoJSONFeatureCollection

Field	Tipe	Status	Keterangan / Validasi
`type`	`string`	Opsional	default=FeatureCollection
`cycle_id`	`integer | null`	Opsional	
`cycle_name`	`string | null`	Opsional	
`features`	`array[GeoJSONFeature]`	**Wajib**	

GeoJSONGeometry

GeoJSON geometry supporting LineString and Polygon.

Field	Tipe	Status	Keterangan / Validasi
`type`	`string (enum: 'LineString', 'Polygon')`	**Wajib**	
`coordinates`	`array[any]`	**Wajib**	

LoginRequest

Field	Tipe	Status	Keterangan / Validasi
`username`	`string`	**Wajib**	minLen=1, maxLen=50
`password`	`string`	**Wajib**	minLen=1

MapSegmentDetailResponse

Field	Tipe	Status	Keterangan / Validasi
`segment_id`	`integer`	**Wajib**	
`segment_code`	`string`	**Wajib**	
`name`	`string`	**Wajib**	
`corridor`	`CorridorEnum`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`geojson_geometry`	`object`	**Wajib**	
`cycle_id`	`integer | null`	Opsional	
`cycle_name`	`string | null`	Opsional	
`ahp_weights`	`object | null`	Opsional	
`indicators`	`object | null`	Opsional	
`normalized_values`	`object | null`	Opsional	
`weighted_values`	`object | null`	Opsional	
`d_plus`	`number | null`	Opsional	
`d_minus`	`number | null`	Opsional	
`preference_score`	`number | null`	Opsional	
`rank`	`integer | null`	Opsional	
`priority_category`	`string`	**Wajib**	
`is_eligible`	`boolean | null`	Opsional	
`ineligible_reason`	`string | null`	Opsional	
`is_tie_broken`	`boolean`	Opsional	default=False
`inspection_date`	`string (date) | string | null`	Opsional	
`data_source`	`string | null`	Opsional	

ObservationBriefResponse

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`segment_id`	`integer`	**Wajib**	
`total_lights`	`integer`	**Wajib**	
`broken_lights`	`integer`	**Wajib**	
`dawn_crime_count`	`integer | null`	**Wajib**	
`boarding_house_count`	`integer`	**Wajib**	
`inspection_date`	`string (date)`	**Wajib**	
`data_source`	`DataSourceEnum`	**Wajib**	
`updated_by`	`integer`	**Wajib**	
`updated_at`	`string (date-time)`	**Wajib**	

ObservationFormRequest

Field	Tipe	Status	Keterangan / Validasi
`segment_id`	`integer | null`	Opsional	
`segment_code`	`string | null`	Opsional	
`total_lights`	`integer`	**Wajib**	min=0.0
`broken_lights`	`integer`	**Wajib**	min=0.0
`dawn_crime_count`	`integer | null`	Opsional	
`boarding_house_count`	`integer`	**Wajib**	min=0.0
`inspection_date`	`string (date)`	**Wajib**	

ObservationFormResponse

Field	Tipe	Status	Keterangan / Validasi
`observation`	`ObservationBriefResponse`	**Wajib**	
`warnings`	`array[string]`	Opsional	

SegmentCreateRequest

Field	Tipe	Status	Keterangan / Validasi
`segment_code`	`string`	**Wajib**	minLen=3, maxLen=50
`name`	`string`	**Wajib**	minLen=3, maxLen=200
`corridor`	`CorridorEnum`	**Wajib**	
`geojson_geometry`	`GeoJSONGeometry`	**Wajib**	
`is_active`	`boolean`	Opsional	default=True

SegmentDetailResponse

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`segment_code`	`string`	**Wajib**	
`name`	`string`	**Wajib**	
`corridor`	`CorridorEnum`	**Wajib**	
`geojson_geometry`	`object`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`updated_at`	`string (date-time)`	**Wajib**	
`observation`	`ObservationBriefResponse | null`	Opsional	

SegmentResponse

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`segment_code`	`string`	**Wajib**	
`name`	`string`	**Wajib**	
`corridor`	`CorridorEnum`	**Wajib**	
`geojson_geometry`	`object`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`updated_at`	`string (date-time)`	**Wajib**	

SegmentUpdateRequest

Field	Tipe	Status	Keterangan / Validasi
`name`	`string | null`	Opsional	
`corridor`	`CorridorEnum | null`	Opsional	
`geojson_geometry`	`GeoJSONGeometry | null`	Opsional	
`is_active`	`boolean | null`	Opsional	

TOPSISCycleCreateRequest

Field	Tipe	Status	Keterangan / Validasi
`cycle_name`	`string`	**Wajib**	minLen=2, maxLen=200
`criteria_directions`	`array[any]`	Opsional	default=['max', 'max', 'max']

TOPSISCycleDetailResponse

Field	Tipe	Status	Keterangan / Validasi
`cycle_id`	`integer`	**Wajib**	
`cycle_name`	`string`	**Wajib**	
`executed_by`	`integer`	**Wajib**	
`ahp_version_id`	`integer`	**Wajib**	
`weights_snapshot`	`object`	**Wajib**	
`status`	`CycleStatus`	**Wajib**	
`created_at`	`string (date-time)`	**Wajib**	
`eligible_count`	`integer`	**Wajib**	
`ineligible_count`	`integer`	**Wajib**	
`eligible_results`	`array[CycleRankingItemResponse]`	**Wajib**	
`ineligible_results`	`array[CycleRankingItemResponse]`	**Wajib**	

TokenResponse

Field	Tipe	Status	Keterangan / Validasi
`access_token`	`string`	**Wajib**	
`token_type`	`string`	Opsional	default=bearer
`user`	`UserResponse`	**Wajib**	

UserCreateRequest

Field	Tipe	Status	Keterangan / Validasi
`username`	`string`	**Wajib**	minLen=3, maxLen=50
`email`	`string (email)`	**Wajib**	
`full_name`	`string`	**Wajib**	minLen=2, maxLen=150
`password`	`string`	**Wajib**	minLen=6, maxLen=128
`role`	`UserRole`	**Wajib**	
`is_active`	`boolean`	Opsional	default=True

UserResponse

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`username`	`string`	**Wajib**	
`email`	`string (email)`	**Wajib**	
`full_name`	`string`	**Wajib**	
`role`	`UserRole`	**Wajib**	
`is_active`	`boolean`	**Wajib**	
`created_at`	`string (date-time)`	**Wajib**	

UserRole

Enum values: `ADMIN`, `OPERATOR`, `PENILAI`, `PENANGGUNG_JAWAB_PJU`

UserUpdateRequest

Field	Tipe	Status	Keterangan / Validasi
`full_name`	`string | null`	Opsional	
`role`	`UserRole | null`	Opsional	
`is_active`	`boolean | null`	Opsional	

WorklistCandidateItem

Field	Tipe	Status	Keterangan / Validasi
`segment_id`	`integer`	**Wajib**	
`segment_code`	`string`	**Wajib**	
`segment_name`	`string`	**Wajib**	
`corridor`	`CorridorEnum`	**Wajib**	
`rank`	`integer`	**Wajib**	
`preference_score`	`number`	**Wajib**	
`broken_lights`	`integer`	**Wajib**	
`dawn_crime_count`	`integer`	**Wajib**	
`boarding_house_count`	`integer`	**Wajib**	
`is_tie_broken`	`boolean`	**Wajib**	

WorklistDecisionResponse

Field	Tipe	Status	Keterangan / Validasi
`id`	`integer`	**Wajib**	
`cycle_id`	`integer`	**Wajib**	
`decided_by`	`integer`	**Wajib**	
`capacity_quota`	`integer`	**Wajib**	
`system_proposed_segment_ids`	`array[integer]`	**Wajib**	
`final_selected_segment_ids`	`array[integer]`	**Wajib**	
`is_overridden`	`boolean`	**Wajib**	
`override_reason`	`string | null`	**Wajib**	
`finalized_at`	`string (date-time)`	**Wajib**	

WorklistFinalizeRequest

Field	Tipe	Status	Keterangan / Validasi
`capacity_quota`	`integer`	**Wajib**	min > 0.0
`final_selected_segment_ids`	`array[integer]`	**Wajib**	
`override_reason`	`string | null`	Opsional	

WorklistProposeRequest

Field	Tipe	Status	Keterangan / Validasi
`capacity_quota`	`integer`	**Wajib**	Kapasitas jumlah gang K > 0 — min > 0.0

WorklistProposeResponse

Field	Tipe	Status	Keterangan / Validasi
`cycle_id`	`integer`	**Wajib**	
`cycle_name`	`string`	**Wajib**	
`capacity_quota`	`integer`	**Wajib**	
`total_eligible`	`integer`	**Wajib**	
`system_proposed`	`array[WorklistCandidateItem]`	**Wajib**	
`reserve_candidates`	`array[WorklistCandidateItem]`	**Wajib**	